

Vector Integration: Advanced Stokes' Theorem Application

1. Stokes' Theorem with a Tilted Planar Intersecting Surface

Name of Concepts

- 1. Stokes' Theorem Line Integral Reduction.
- 2. Plane Projection of a Tilted Elliptical Intersection Region.
- 3. Area of an Inclined Plane Projection via 2D Ellipse Mapping.

Core Mathematical Formulas

Stokes' Theorem vector equivalence mapping statement:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

For a planar surface equation given in standard form $Ax + By + Cz + D = 0$, the constant unit outward normal vector $\hat{n}$ is:

$$\hat{n} = \frac{A\vec{i} + B\vec{j} + C\vec{k}}{\sqrt{A^2 + B^2 + C^2}}$$

The standard geometric area formula for an orthogonal 2D ellipse with semi-axes lengths a and b is:

$$\text{Area} = \pi ab$$

Beginner-Friendly Explanation of Operations

- 1. **Completing the Fragmented Question Context:** The raw prompt text omits the second bounding constraint equation. Based on standard vector examination curriculum patterns for this exact vector problem, the missing path limit is the tilted plane equation:
$$z = x + 3$$
- 2. **Visualizing the Intersecting Region:** Imagine a perfect sphere centered at a height on the z-axis. Now imagine a flat blade cutting through this sphere at a tilted angle along the line $z = x + 3$. The perimeter rim where the blade shears the skin of the sphere forms an inclined ellipse path in 3D space, which is our curve C .
- 3. **Choosing the Surface (S):** Stokes' Theorem allows us to choose *any* surface that is capped by the boundary loop C . Instead of picking the bulging curved skin of the sphere, the easiest choice is the perfectly flat, tilted disk inside the plane $z = x + 3$ itself.
- 4. **The Constant Vector Trick:** Because our chosen surface S is completely flat, its orientation never changes. This means its normal perpendicular vector $\hat{n}$ is a constant vector across the entire region. When we compute the dot product $(\nabla \times \vec{F}) \cdot \hat{n}$, it results in a constant scalar number. A constant scalar number can be pulled completely out of the integral loop, simplifying our surface calculation to a basic geometry problem:

$$\iint_S (\text{Constant}) dS = \text{Constant} \times (\text{Total Surface Area of the Tilted Disk } S)$$

2. Step-by-Step Problem Solution

Question

Apply Stokes' theorem to calculate $\oint_C 4y \, dx + 2z \, dy + 6y \, dz$, where C is the curve of intersection of the sphere $x^2 + y^2 + z^2 = 6z$ and the plane $z = x + 3$.

Step 1: Identify components and verify the Curl vector ($\nabla \times \vec{F}$)

Extract the components of the force vector field:

$$f_1 = 4y, \quad f_2 = 2z, \quad f_3 = 6y$$

Set up the matrix determinant to calculate the rotation vector field:

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4y & 2z & 6y \end{vmatrix}$$

Expand the components step-by-step:

$$\text{Component } \vec{i} = \left[\frac{\partial}{\partial y}(6y) - \frac{\partial}{\partial z}(2z) \right] = 6 - 2 = 4$$

$$\text{Component } \vec{j} = - \left[\frac{\partial}{\partial x}(6y) - \frac{\partial}{\partial z}(4y) \right] = -(0 - 0) = 0$$

$$\text{Component } \vec{k} = \left[\frac{\partial}{\partial x}(2z) - \frac{\partial}{\partial y}(4y) \right] = 0 - 4 = -4$$

Assemble the calculated vector:

$$\nabla \times \vec{F} = 4\vec{i} - 4\vec{k}$$

Step 2: Compute the unit normal vector ($\hat{n}$) of the planar surface

Rearrange the planar cutting constraint to isolate the constant coefficients:

$$z = x + 3 \implies -x + 0y + z - 3 = 0$$

Find the gradient direction vector components by extracting the coefficients of x, y, z :

$$\vec{N} = -\vec{i} + 0\vec{j} + \vec{k}$$

Normalize the vector by dividing by its geometric length to create a unit vector:

$$|\vec{N}| = \sqrt{(-1)^2 + (0)^2 + (1)^2} = \sqrt{1 + 0 + 1} = \sqrt{2}$$
$$\hat{n} = \frac{-\vec{i} + \vec{k}}{\sqrt{2}}$$

Step 3: Compute the scalar dot product $(\nabla \times \vec{F}) \cdot \hat{n}$

Multiply corresponding coordinate components together:

$$(\nabla \times \vec{F}) \cdot \hat{n} = (4\vec{i} - 4\vec{k}) \cdot \left(\frac{-1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{k} \right)$$
$$(\nabla \times \vec{F}) \cdot \hat{n} = \left(4 \cdot \frac{-1}{\sqrt{2}} \right) + (0 \cdot 0) + \left(-4 \cdot \frac{1}{\sqrt{2}} \right)$$
$$(\nabla \times \vec{F}) \cdot \hat{n} = \frac{-4}{\sqrt{2}} - \frac{4}{\sqrt{2}} = \frac{-8}{\sqrt{2}}$$

Simplify the radical fraction value:

$$\frac{-8}{\sqrt{2}} = \frac{-4 \cdot 2}{\sqrt{2}} = -4\sqrt{2}$$

Step 4: Substitute the constant value back into Stokes' template

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (-4\sqrt{2}) dS$$

Pull the constant factor outside the double integral limits:

$$\oint_C \vec{F} \cdot d\vec{r} = -4\sqrt{2} \iint_S dS = -4\sqrt{2} \times (\text{Area of Surface } S)$$

Step 5: Find the boundary shape of the tilted surface area

Substitute the plane link condition $z = x + 3$ directly into the sphere's equation to project the 3D tilted boundary down onto the flat 2D xy -plane:

$$x^2 + y^2 + (x + 3)^2 = 6(x + 3)$$

Expand the squared binomial expressions:

$$x^2 + y^2 + (x^2 + 6x + 9) = 6x + 18$$

Combine like algebraic elements and shift constants to the right side:

$$2x^2 + y^2 + 6x + 9 = 6x + 18$$

Subtract $6x$ from both sides:

$$2x^2 + y^2 + 9 = 18$$

$$2x^2 + y^2 = 9$$

Step 6: Identify the 2D projected boundary dimensions

Convert the equation to the standard ellipse format by dividing all terms by 9:

$$\frac{2x^2}{9} + \frac{y^2}{9} = 1 \implies \frac{x^2}{(9/2)} + \frac{y^2}{9} = 1$$

Extract the semi-major and semi-minor scaling radius boundaries:

$$a^2 = \frac{9}{2} \implies a = \frac{3}{\sqrt{2}}$$

$$b^2 = 9 \implies b = 3$$

Calculate the flat 2D area projected onto the floor (xy -plane):

$$\text{Projected Area } (A_{xy}) = \pi \cdot a \cdot b = \pi \left(\frac{3}{\sqrt{2}} \right) (3) = \frac{9\pi}{\sqrt{2}}$$

Step 7: Calculate the actual 3D area of the tilted surface S

When an area is tilted at an angle, its shadow projection onto the floor shrinks. To find the true, un-skewed 3D area of the tilted disk, we divide the flat shadow area by the cosine of the tilt angle ($\cos \gamma$, where γ is the angle between normal vector $\hat{n}$ and vertical unit axis $\bar{k}$):

$$\cos \gamma = \hat{n} \cdot \bar{k} = \left(\frac{-1}{\sqrt{2}} \bar{i} + \frac{1}{\sqrt{2}} \bar{k} \right) \cdot \bar{k} = \frac{1}{\sqrt{2}}$$

$$\text{True Area of } S = \frac{\text{Projected Area } (A_{xy})}{\cos \gamma} = \frac{\left(\frac{9\pi}{\sqrt{2}} \right)}{\left(\frac{1}{\sqrt{2}} \right)} = 9\pi$$

Step 8: Complete the final scalar evaluation

Substitute the calculated surface area value back into our simplified Stokes' equation from **Step 4**:

$$\oint_C \vec{F} \cdot d\vec{r} = -4\sqrt{2} \times (9\pi)$$

$$\oint_C \vec{F} \cdot d\vec{r} = -36\sqrt{2}\pi$$